Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy of MLP by dataset.

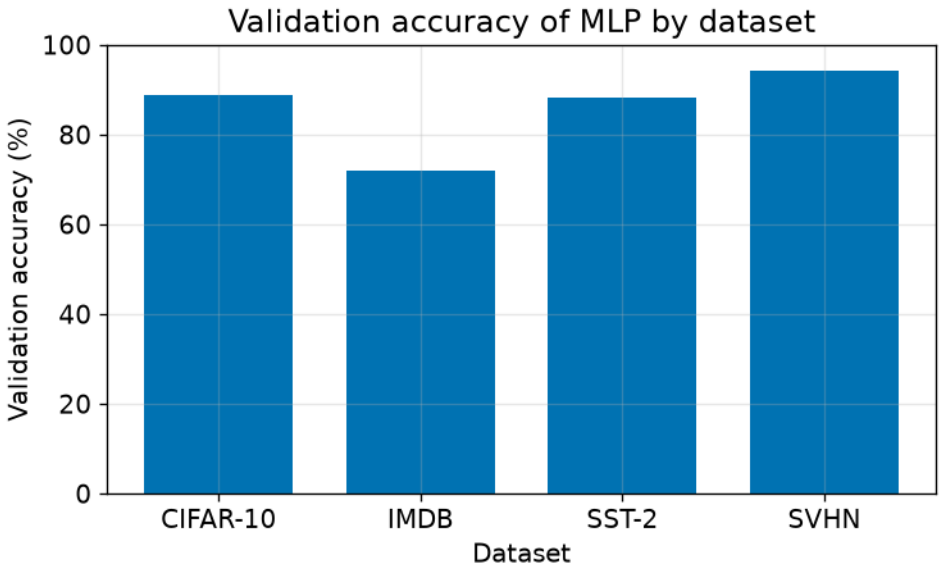


Figure 1: Validation accuracy of MLP by dataset for the models in this experiment.

On IMDB, MLP obtains a validation accuracy of 122.1%. MLP achieves 77.1% validation accuracy on IMDB.

Figure 2 shows the training loss.

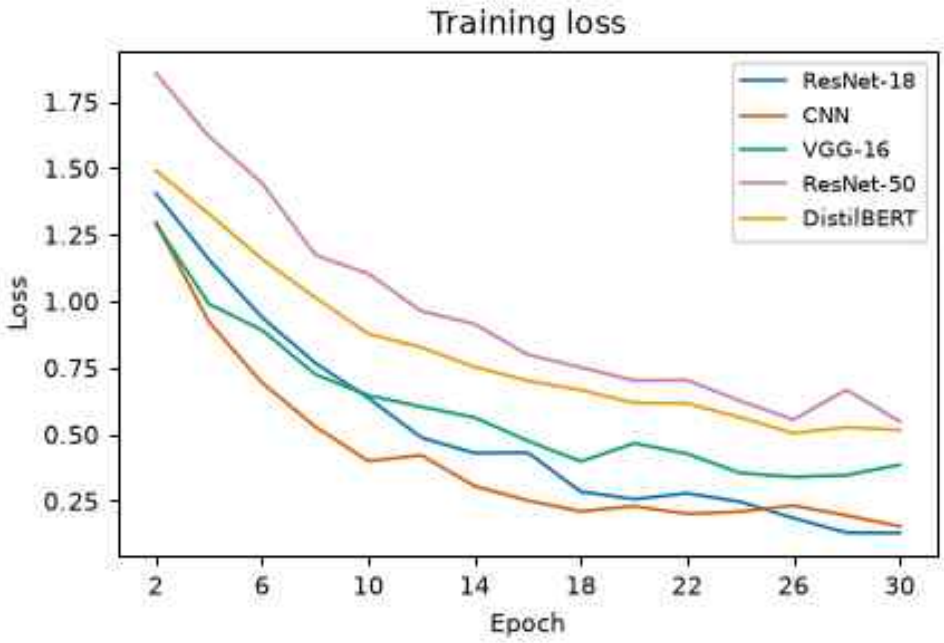


Figure 2: Training loss for the models in this experiment.

CNN has a loss of 0.731 at epoch 4. At epoch 22, the loss of VGG-16 is 0.389 (Figure 2). At epoch 26, the loss of ResNet-18 is 0.185. At epoch 8, the loss of VGG-16 is 1.105.

Figure 3 shows the training loss.

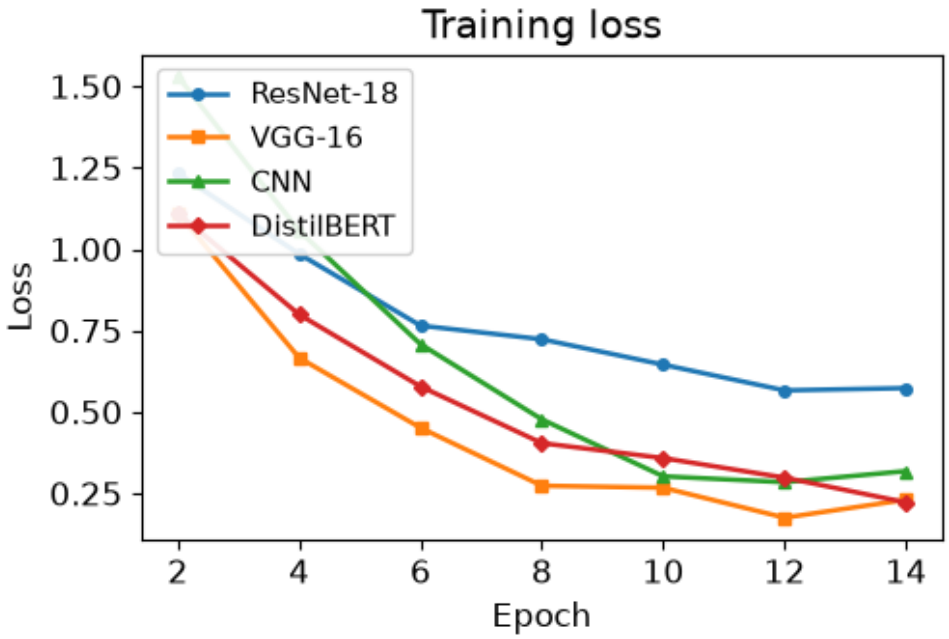


Figure 3: Training loss for the models in this experiment.

At epoch 10, the loss of CNN is 0.358. DistilBERT has a loss of 1.110 at epoch 2. ResNet-18 has a loss of 0.267 at epoch 10 (Figure 3).

3 Discussion

Random Forest reaches 69.3% accuracy on QQP. Overall, the results are in line with what we expected from the literature.